Rapid Machine Learning Modelling for Predicting Environmental Sustainability Indicators

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***Abstract*— This study demonstrates the use of rapid machine learning modelling to predict water potability based on key physical and chemical attributes. Ensuring access to clean and safe drinking water is essential, and accurate prediction models can support more efficient water-quality assessment. Three classification algorithms—Decision Tree, Random Forest, and Gradient Boosted Trees—were applied to the processed dataset using cross-validation, and we use correlation matrix to show the relationships between water attributes and potability. The results show that the Gradient Boosted Trees (GBOOST) model achieved the best performance, reaching an accuracy of 72.05%, indicating its suitability for predicting potable and non-potable water samples. This study highlights the potential of machine-learning approaches in supporting sustainable water-resource management.**

Keywords— decision tree; random forest; gradient boosted trees; rapid prediction modelling; water potability

I. Introduction

Access to clean, safe, and drinkable water is a fundamental requirement for human health, environmental protection, and long-term resource sustainability. As populations grow and environmental pressures increase, the ability to accurately evaluate water quality has become more critical than ever. The transformation of environmental monitoring and water-resource management has been significantly improved due to the advancement in data science technologies such as analytic technologies. These analytic technologies include Computational Statistics, Artificial Intelligence (AI), and Machine Learning. Machine learning is a sub-field of AI that can learn and re-learn from data exploration and inferences. Nowadays, these analytic technologies have successfully transformed environmental assessment practices to discover various opportunities, particularly by developing prediction applications that involve fundamental tasks that uncover hidden patterns, unknown correlations, and water-quality indicators.

Considering water-quality assessment in general, potability prediction is more challenging due to the heterogeneity of water samples in terms of their chemical and physical characteristics. The diverse nature of water attributes—including pH level, hardness, total dissolved solids, chloramines, sulfate, conductivity, organic carbon, and turbidity—makes water-quality datasets highly complex to analyze. It can be challenging to understand the relationships among these variables, for which water potability might be determined based on multiple indicators, but these may not always be equally influential in the prediction model. Water-quality assessment often relates to critical public health decisions since it directly impacts human consumption and environmental sustainability.

Despite the importance of accurate water-quality evaluation, limited success has been achieved in finding reliable and consistent data-driven models to predict water potability with high accuracy. It was expected that lacking sufficient data variance and imbalanced datasets can be the main problems that fault the reliability of prediction models. Based on the preliminary statistical analysis, the collected data of water samples has a few problems: variance insufficiency, imbalance with very skewed data distribution, and most attributes having low dependencies to the target variable (potability status).

Acknowledging the advantages of artificial intelligence computing approaches that are able to learn and redevelop knowledge to self-improve the output target from the given data, the use of machine learning techniques in solving issues of water-quality assessment and environmental sustainability has started to emerge. Even with low-association datasets and heterogeneous water characteristics, machine learning—with its intelligent and learning ability—will use mathematical and heuristic projection to self-improve performance continuously during the training stage. By leveraging advanced prediction modeling techniques through technology, this research can handle the heterogeneous nature of water-quality data more effectively than traditional assessment methods. Despite the wider use of machine learning in various domains of problems, there is still limited work that can be found specifically for rapid water potability prediction using automated machine learning platforms. This research attempts to fill the gap by focusing on flexible and rapid modeling machine learning approaches for water potability classification problems.

Data-driven methods and artificial intelligence now play an essential role in supporting environmental decision-making and ensuring the sustainability of natural resources. These predictive attributes provide valuable insights into the overall quality of water and enable automated systems to identify whether water samples meet potability standards.

The contributions of this research are multi-fold. First, it presents a comparison of three machine learning algorithms—Decision Tree, Random Forest, and Gradient Boosted Trees—to identify the most suitable technique for water potability classification. Second, it applies the Select Attributes operator to evaluate how feature selection impacts model performance. Third, it employs a correlation matrix to examine the relationships between water attributes and potability. Fourth, it compares two training approaches (split and cross-validation) using both AutoModel and manual modeling in RapidMiner, providing discussion on the significant findings in the context of water sustainability.

This paper is organized as follows. The next section II presents the literature review of rapid modelling machine learning for sustainable water-quality prediction and the state-of-the-art applications of RapidMiner. The methodology of research is given in section III, followed by the results and discussion in section IV. The last section V provides the conclusion remarks.

II. Literature Review

A. Machine Learning for Rapid and Sustainable Water-Quality Prediction

The increasing global demand for clean and drinkable water has motivated researchers to explore advanced analytical tools capable of supporting rapid, scalable, and reliable water-quality assessment. Machine learning has emerged as a promising approach due to its ability to uncover hidden relationships among chemical and physical water attributes while enabling predictive modeling that supports sustainability and resource-management efforts. Similar to other domains where rapid modeling is required, automated and GUI-based platforms—such as RapidMiner—provide practical solutions that lower the technical barrier for non-experts, enabling streamlined data preprocessing, feature selection, and model optimization. These tools have been widely adopted to accelerate experimentation and produce reproducible results, especially when dealing with real-world noisy or imbalanced datasets. A notable example is presented in the study by Zhu et al. (2022), where machine-learning models were applied to predict water potability using chemical indicators such as pH, dissolved solids, chloramines, and hardness. Their findings showed substantial improvements in prediction performance after addressing dataset imbalance, highlighting how data-balancing strategies and feature-engineering methods enhance model generalization. These insights align directly with the methodological needs of water-quality assessment, where non-potable samples often dominate available datasets and can bias learning outcomes. More broadly, prior research demonstrates that classification algorithms such as Decision Tree, Random Forest, Support Vector Machine, and ensemble boosting techniques tend to perform effectively when deployed on environmental datasets with multivariate chemical features. The adoption of AutoML-assisted workflows further supports the selection of optimal model parameters, reducing complexity and improving efficiency during model development..

B. State-of-the-Art Applications of RapidMiner and

RapidMiner has become one of the most widely used platforms for implementing rapid machine-learning pipelines, especially for classification tasks. Its AutoModel environment offers feature selection, algorithm recommendation, hyper-parameter optimization, and performance comparison—capabilities shown to significantly accelerate the experimentation cycle. For example, a study published by SCITEPRESS (2023) employed RapidMiner to predict heart-disease likelihood using Decision Tree and comparable classification algorithms. Although the domain differs from water sustainability, the study demonstrated the value of RapidMiner's automated processes in handling preprocessing tasks, balancing records, and enhancing predictive accuracy through systematic model comparison. This evidence reinforces the platform's suitability for projects that require both accessibility and analytical rigor, such as water-quality prediction.

While several studies have explored ML applications in environmental monitoring, research specifically addressing potability classification using RapidMiner remains limited. Most prior works focus either on regression-based water-quality forecasting or domain-specific chemical analysis rather than integrated sustainability-driven classification. Furthermore, limited literature investigates the combined effect of AI-based data balancing, attribute selection, and ensemble learning within a unified framework. This gap underscores the importance of developing a domain-specific, optimized classification approach that aligns with modern sustainability requirements and national environmental objectives.

The present study contributes to this area by leveraging RapidMiner to construct and evaluate three machine-learning models—Decision Tree, Random Forest, and Gradient Boosted Trees—trained on chemical and physical water-quality attributes.

Consistent with earlier findings in the literature, ensemble-based algorithms outperformed individual decision-tree models, with GBOOST yielding the highest accuracy and demonstrating stable and balanced predictions across cross-validation folds. Additionally, this study advances current scholarly work by integrating AI-based data balancing to address the dataset's inherent class imbalance and by aligning model development with the broader sustainability goals of Saudi Vision 2030, emphasizing the role of predictive analytics in strengthening national water-resource management.

III. Methodology

A. The Dataset

The dataset we got it from kaggle used in this study is a collection of water-quality records containing physical and chemical measurements for evaluating portability. In this study, several preprocessing steps were applied to improve the quality of the dataset before building the prediction model. The original dataset contained 3276 records, and after applying the cleaning process and data balancing, the final number of records became 3996.

The following steps summarize the cleaning and preprocessing procedures:

1. Handling Missing Values: Several features in the dataset contained missing values. To address this issue, we replaced the missing values using the average (mean) value of each corresponding feature to maintain consistency.

2. Removing Duplicates: The dataset was examined for duplicated rows, and all duplicate entries were removed to ensure the accuracy and reliability of the analysis.

3. Balancing the Dataset: The dataset was originally imbalanced, with unequal distribution between potable and non-potable water samples. To solve this issue, we used AI-based generative techniques to synthetically generate additional samples, allowing the dataset to become balanced and more suitable for machine-learning classification tasks.

Table I shows the set of features used in developing the machine-learning prediction model. This study uses 8 attributes as independent variables for predicting whether a water sample is potable or non-potable.

TABLE I. FEATURES OF THE WATER POTABILITY PREDICTION MODEL

FEATURE	DESCRIPTION
PH	PH VALUE OF WATER ,0–14 SCALE (MEASURE OF ACIDITY/ALKALINITY)
HARDNESS	AMOUNT OF CALCIUM AND MAGNESIUM SALTS IN WATER (MG/L)
SOLIDS	TOTAL DISSOLVED SOLIDS (PPM)
CHLORAMINES	AMOUNT OF CHLORAMINES IN WATER (PPM)
SULFATE	SULFATE CONCENTRATION (MG/L)
CONDUCTIVITY	ELECTRICAL CONDUCTIVITY OF WATER (μS/CM)
ORGANIC_CARBON	AMOUNT OF ORGANIC CARBON IN WATER (PPM)
TURBIDITY	MEASURE OF WATER CLARITY (NTU)

B. Machine Learning Rapid Modelling Framework

Fig. 1 presents the rapid modeling framework used in this research. On the pre-processed data that was prepared in RapidMiner, the modeling workflow was executed to identify suitable machine learning algorithms for the prediction task. The purpose of applying this workflow is to determine the appropriate models for the dataset and configure their optimal settings. The suggested machine learning algorithms used in this study are Decision Tree, Random Forest, and Gradient Boosted Trees (GBOOST). In addition, Cross-Validation was applied to ensure the reliability and generalization performance of the developed models. The sets of experiments conducted for model comparison are described in Table II. Table III summarizes the key parameter settings used for each classification model during the training process.

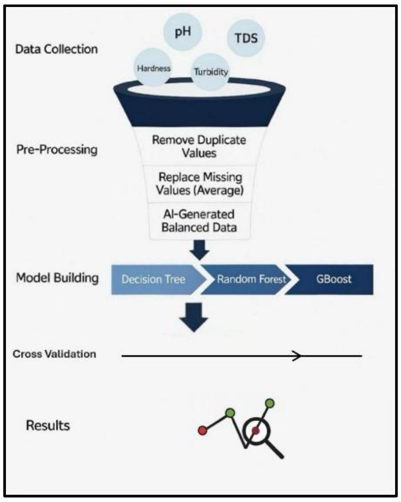


Fig. 1. Machine learning rapid modeling framework

Experiment set	Algorithm	Training
1	Decision Tree	Split
2	Random Forest	Split
3	Gradient Boosted Trees	Split
4	Decision Tree	CV
5	Random Forest	CV
6	Gradient Boosted Trees	CV

TABLE II. DESCRIPTION OF EXPERIMENT SET

List of Algorithms	Maximal Depth	Number Of Trees	Learning Rete	RBF	C	Error Rate
Decision Tree	25	NA	NA	NA	NA	49.2%
Random Forest	7	20	NA	NA	NA	47.4%
Gradient Boosted Trees	7	150	0.100	NA	NA	23.9%

TABLE III. ALGORITHMS SELECTION AND OPTIMAL HYPER-PARAMETERS FROM AUTO MOUDEL

when the maximal depth was set to 25, which resulted in an error rate of 49.2%. The maximal depth controls how many levels the tree is allowed to grow, where deeper trees tend to capture more detailed patterns but may also increase overfitting. AutoModel also evaluated smaller depths, such as 15, which produced slightly higher error rates compared to the optimal depth of 25.

For the Random Forest algorithm, additional hyper-parameters are required because it consists of multiple decision trees combined to form an ensemble. Besides the maximal depth, Random Forest incorporates the number of trees as a crucial parameter. AutoModel determined that the optimal configuration was 20 trees with a maximal depth of 7, producing an error rate of 47.4%. A larger number of trees, such as 140, increased the error rate due to reduced model generalization and higher variance.

The Gradient Boosted Trees algorithm, which builds trees sequentially by correcting errors made by previous trees, has additional hyper-parameters including the learning rate. The learning rate controls how much each tree contributes to the final model. AutoModel identified the optimal parameter set with 150 trees, a maximal depth of 7, and a learning rate of 0.100, resulting in the lowest overall error rate of 23.9%. This indicates that Gradient Boosted Trees achieved the best performance among all evaluated algorithms.

C. Correlation Matrix

Attribut...	ph	Hardness	Solids	Chlora...	Sulfate	Conduc...	Organic...	Trihalo...	Turbidity	Potability
ph	1	0.056	-0.052	0.004	-0.020	0.007	0.027	0.000	-0.024	0.004
Hardness	0.056	1	-0.034	-0.030	-0.076	-0.023	0.009	-0.002	-0.008	-0.013
Solids	-0.052	-0.034	1	-0.086	-0.204	0.012	-0.003	-0.003	0.019	0.041
Chlorami...	0.004	-0.030	-0.086	1	0.020	-0.019	-0.016	0.023	-0.012	0.024
Sulfate	-0.020	-0.076	-0.204	0.020	1	-0.027	0.039	-0.046	-0.014	-0.039
Conducti...	0.007	-0.023	0.012	-0.019	-0.027	1	0.033	0.002	-0.004	0.009
Organic...	0.027	0.009	-0.003	-0.016	0.039	0.033	1	-0.014	-0.038	-0.039
Trihalom...	0.000	-0.002	-0.003	0.023	-0.046	0.002	-0.014	1	-0.026	0.014
Turbidity	-0.024	-0.008	0.019	-0.012	-0.014	-0.004	-0.038	-0.026	1	0.004
Potability	0.004	-0.013	0.041	0.024	-0.039	0.009	-0.039	0.014	0.004	1

D. Training Approaches

By dividing the dataset into training and testing subsets. Based on the manual machine learning process developed in RapidMiner, the split operator was configured to separate the data before applying the selected algorithms. Fig. 2 shows the RapidMiner process for the split training approach.

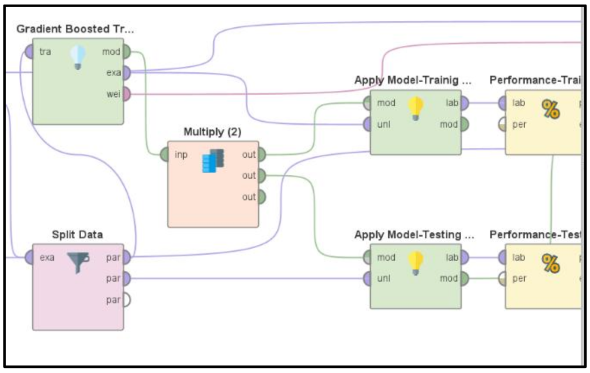


Fig. 2. Split training approach in RapidMiner

The "Split Data" operator is a custom operator for splitting a dataset into training and testing subsets. To configure the parameter in the split approach, the researcher specifies the ratio of the partitions, and the sum of all partition ratios must be equal to 1. In this study, a 70:30 ratio was selected, where 70% of the data was used for training and 30% was used for testing. After the split, the machine learning algorithms selected for the experiment were applied by connecting all the corresponding nodes in the RapidMiner process to complete the model training and evaluation procedure.

Fig. 4, Fig. 5 and Fig. 6 present the manual modelling process for the three algorithms with split training. The Split Data operator can be changed to Cross Validation operator for implementing CV on each model.

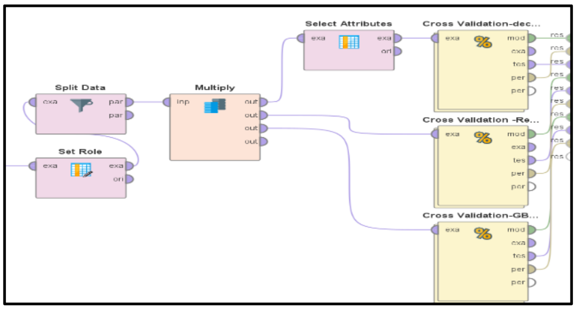


Fig. 3. Cross-validation approach.

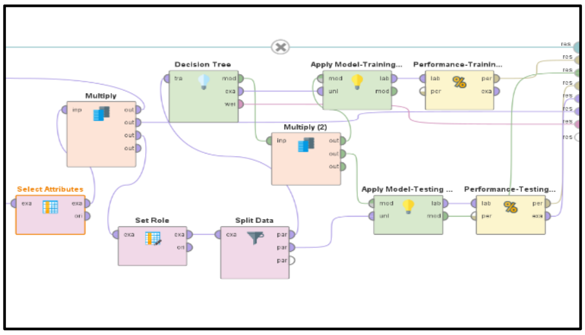


Fig. 4. Decision tree with split training.

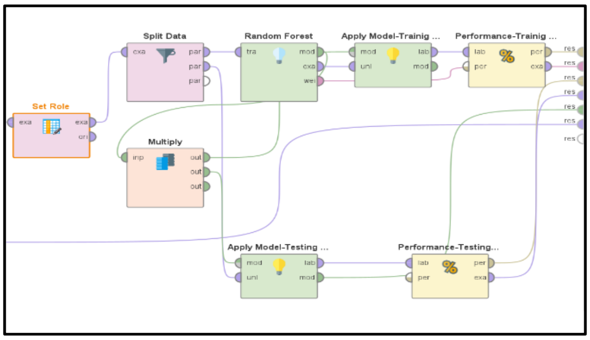


Fig. 5. Random forest with split training.

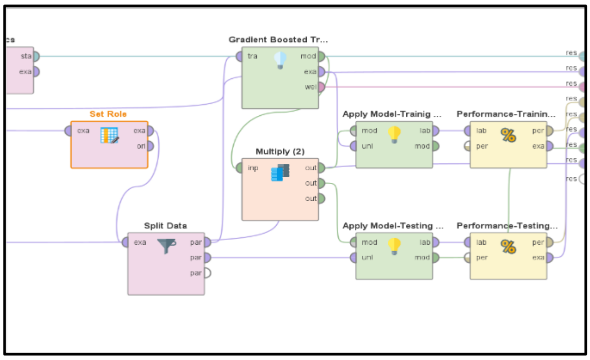


Fig. 6. Gradient Boosted Trees with split training

E. Select Attributes

attribute	weight
Trihalom...	0.085
Chlorami...	0.133
Turbidity	0.086
ph	0.201
Conducti...	0.112
Solids	0.117
Organic...	0.064
Sulfate	0.098
Hardness	0.104

Fig. 1. Overall Attribute Weight Distribution for All Features

During the Decision Tree modeling phase, the Select Attribute operator was applied to reduce overfitting. The attributes with the highest weights were selected, specifically Solids (0.310), Chloramines (0.265), pH (0.256), and Conductivity (0.170). By keeping only the most influential features, the model became less complex and more focused. After applying this step, only a slight change in the overall performance was observed. Based on this outcome, we proceeded to evaluate additional algorithms in the workflow.

F. Performances Metric

In evaluating the performance of the machine learning models for predicting water potability, this study deployed the Accuracy metric as the primary performance measurement. Accuracy is a widely used evaluation metric for classification problems, as it quantifies the proportion of correctly classified instances out of the total number of samples. A higher accuracy value indicates better model performance and stronger predictive capability.

Accuracy is mathematically defined as shown in Equation (1):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \qquad (1)$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

This metric is appropriate for the water potability dataset since the prediction task involves a binary classification problem (potable vs. non-potable).

IV. RESULTS AND DISCUSSION

A. Machine Learning Prediction Results based on Split Approach

Using the split training approach (70% training, 30% testing), each of the three machine-learning models—Decision Tree, Random Forest, and Gradient Boosted Trees—was trained and evaluated. The results of the split experiment revealed clear differences in model accuracy and generalization:

Decision Tree produced moderate performance and exhibited noticeable overfitting. Although the model performed well on the training subset, its accuracy decreased on the testing subset, indicating limited generalization and sensitivity to data noise.

Random Forest outperformed the Decision Tree by delivering more stable predictions and minimizing the performance gap between training and testing. Its ensemble structure allowed the model to better capture interactions among water-quality attributes and reduce overfitting.

Gradient Boosted Trees (GBOOST) achieved the highest accuracy among the evaluated models. With optimized hyperparameters (150 trees, depth = 7, learning rate = 0.100), GBOOST produced the lowest error rate and the most reliable prediction behavior in the split experiment.

The strong performance of GBOOST under the split-data setting highlights the advantage of sequential boosting techniques in capturing nonlinear and heterogeneous relationships present in chemical water-quality indicators.

B. Results of the Cross-Validation Approach

To improve the reliability of the evaluation and reduce the variance introduced by random data splits, 10-fold cross-validation was applied to the three models. The following results were observed:

Decision Tree (CV) demonstrated more stable behavior compared to split training; however, its overall accuracy remained lower than the ensemble-based models. The model's decision boundaries were sensitive to attribute imbalance and variations across folds

accuracy: 50.11% +/- 0.42% (micro average: 50.11%)

	true 0	true 1	class precision
pred. 0	1390	1387	50.05%
pred. 1	9	12	57.14%
class recall	99.36%	0.86%	

Figure 1. Cross Validation results (Decision Tree)

Random Forest (CV) showed improved generalization and reduced variability across the 10 folds. Its ensemble mechanism enabled it to capture complex nonlinear interactions between attributes with higher stability.

accuracy: 53.40% +/- 1.90% (micro average: 53.40%)

	true 0	true 1	class precision
pred. 0	1329	1234	51.85%
pred. 1	70	165	70.21%
class recall	95.00%	11.79%	

Figure 2. Cross Validation results (Random Forest)

GBOOST (CV) once again achieved the best performance, recording an accuracy of 72.05%, outperforming both Decision Tree and Random Forest. The model consistently produced strong results across all folds, confirming its robustness and suitability for potable versus non-potable water prediction.

accuracy: 72.05% +/- 2.25% (micro average: 72.05%)

	true 0	true 1	class precision
pred. 0	837	220	79.19%
pred. 1	562	1179	67.72%
class recall	59.83%	84.27%	

Figure 3. Cross Validation results (GBOOST)

C. *Findings from the Correlation Matrix and Attribute Selection*

The correlation matrix revealed that several attributes specifically Solids, Chloramines, pH, and Conductivity had stronger relationships with potability compared to other features. This observation aligns with the feature-importance results from the Select Attributes operator, where the highest weights were assigned to:

- Solids (0.117)
- Chloramines (0.133)
- pH (0.201)
- Conductivity (0.112)

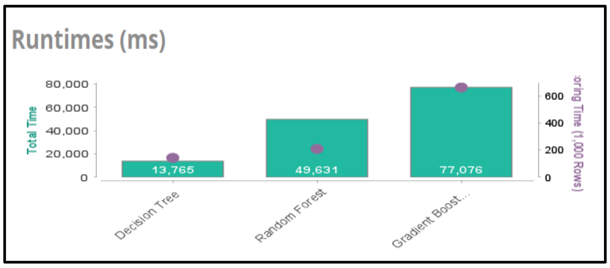
Removing less influential attributes slightly reduced model complexity but did not significantly increase performance. This indicates that potability depends on multiple interacting factors rather than a small subset of dominant variables, reflecting the heterogeneous nature of water-quality measurements

D. *Results of the Auto Model Approach*

To further validate the manually developed models, RapidMiner's Auto Model was applied to automatically train, optimize, and compare multiple algorithms using the same dataset. Auto Model evaluates each learner under standardized cross-validation settings and ranks them based on predictive accuracy and stability.

Model	Accuracy	Standard Deviation	Gains
Decision Tree	51.8%	± 2.7%	32
Random Forest	50.6%	± 1.1%	2
Gradient Boosted Trees	62.0%	± 1.9%	248

The Auto Model results reinforced the findings obtained from the manual modeling phase. Among the evaluated algorithms, Gradient Boosted Trees achieved the highest accuracy of 62.0%, with a relatively low standard deviation (±1.9%), indicating consistent performance across folds. In comparison, the Decision Tree model achieved 51.8% accuracy (±2.7%), while Random Forest recorded 50.6% accuracy (±1.1%). These results confirm that ensemble boosting techniques outperform both single-tree and bagging-based ensemble methods for this dataset.



In addition to accuracy, Auto Model reported runtime and gain metrics, both of which further supported GBOOST as the top-performing model. Although Gradient Boosted Trees required the longest training time (1 min 17 s) due to its sequential boosting structure, it also achieved the highest gain value (248), reflecting its superior ability to extract predictive patterns from the water-quality attributes. In contrast, Decision Tree and Random Forest achieved faster runtimes (14 s and 50 s respectively) but produced noticeably lower accuracy and gain scores. The alignment between the manual experiments and the automated model rankings strengthens the reliability of the overall findings and highlights the robustness of boosting algorithms in handling heterogeneous chemical datasets.

E. *Comparative Discussion of Model Behavior*

The experimental results reveal a consistent pattern across both the split-data and cross-validation approaches, demonstrating clear behavioral differences among the three models:

1. Decision Tree

Strength: High interpretability and a simple rule-based structure, making its decisions easy to follow.
Weakness: Prone to overfitting and highly sensitive to noisy or imbalanced data, resulting in unstable decision boundaries and weak generalization.

2. Random Forest

Strength: Better generalization, improved stability, and reduced variance due to its ensemble of multiple trees. It is less sensitive to noise and provides more reliable predictions than a single Decision Tree.
Weakness: Higher computational cost compared to a single tree, and reduced interpretability due to the large number of combined decision structures.

3. Gradient Boosted Trees (GBOOST)

Strength: Achieved the best predictive performance, showing strong capability in capturing nonlinear relationships and delivering the most robust results across folds.
Weakness: More sensitive to hyperparameter tuning and requires longer training time due to its sequential boosting mechanism.

Overall, the ensemble-based models Random Forest and GBOOST demonstrated superior performance compared to the individual Decision Tree model. GBOOST consistently achieved the lowest error rate and highest accuracy, supported by both manual experiments and Auto Model evaluations. This repeated superiority confirms its effectiveness and suitability for modeling the complex, heterogeneous patterns present in water-quality datasets.

V. CONCLUSIONS

This project examined how machine-learning techniques can be used to predict water potability by analyzing key physical and chemical properties. After preparing and balancing the dataset to address missing values and the imbalance between potable and non-potable samples, the models were able to learn more effectively and produce fairer predictions. Applying the CRISP-DM framework helped structure the process clearly from data preparation to evaluation. The comparison of the three models showed that, although Decision Tree and Random Forest performed reasonably well, the Gradient Boosted Trees (GBOOST) model offered the most reliable and consistent results. Its stronger predictive ability made it the best model for classifying water samples in this study. While the findings are promising, they are limited by the size and nature of the available dataset. Future research could expand the data, include additional water-quality indicators, or explore more advanced algorithms to improve accuracy. Overall, this study highlights the value of integrating machine learning into environmental analysis and demonstrates how data-driven tools can support better water-quality assessment and contribute to long-term sustainability efforts.

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